Statistics for Data Insights



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With assistance from AI tools, particularly Microsoft Copilot and Google Gemini

Outline

- Introduction to Data Analytics
  - Question 1 - What type of data are you working with?
  - Question 2 - What do you want to communicate?
  - Question 3 - Who is the audience and what do they need?
- Measures of Central Tendency
- Measures of Dispersion
- Measures of Distribution Shape
- Analysis of Relationships Between Variables

Introduction to Data Analytics

What is data analytics? 📊

Data analytics = ask a clear question → let the data answer it → say what to do next.

Four questions data can answer: describe → explain → predict → decide - and where each one is covered in this course

📋 Descriptive

✅ OUR FOCUS - 2 LECTURES

"What happened?"

COVERED BY

- **Statistics for Data Insights**

Summaries, distributions, comparing groups.

- **Data Visualization**

Chart choice, plotting and data storytelling.

🔎 Diagnostic

LATER IN THE COURSE

"Why did it happen?"

COVERED BY

- **Probability, Sampling & Uncertainty**
  Select a sample and draw reliable conclusions about the population.
- **Testing, Estimation & A/B Testing**
  Is the observed difference real or random variation (noise)?
- **Causal Inference & Experiments**
  Determine whether an action caused the outcome.

🔮 Predictive

LATER IN THE COURSE

"What is likely next?"

COVERED BY

- **Regression & Classification**

Predict a number or a category.

- **Clustering & Dimensionality Reduction**

Find hidden groups and structure.

🧭 Prescriptive

NOT COVERED

"What should we do?"

- Optimization, simulation, decision models that recommend the best action.

Descriptive analytics

Descriptive Analytics = Statistical Summaries + Data Visualization

🔢 Summarize → 📈 Visualize → 🔍 Interpret → ✅ Recommend

LECTURE 1

Statistics for Data Insights

Discover what the numbers are really telling you and find the story behind the numbers

- 📊 Summary statistics - mean vs median, spread
- 📉 Distributions - shape, skew, outliers
- 👥 Comparing groups and relationships
- 💡 From a raw number to a stated insight

LECTURE 2

Data Visualization

Visualize data to reveal insights for decision-making

- 🧭 Choosing a chart by purpose and audience
- 🎨 Building charts with `Seaborn` and `Plotly`
- 🗣️ Storytelling - headline, evidence, action

3 key questions before any analysis 🧭

1 What type of data are you working with?

Categorical? Numeric? Date-time? Boolean?

📊 STATISTICS Counts and proportions for categories; mean, median and spread for numbers.

📈 VISUALIZATION Bar or stacked bar for categories; histogram or line for numbers over time.

2 What do you want to communicate?

Comparison? Composition? Relationship? Distribution? Trend?

📊 STATISTICS **Group difference** → compare means; **relationship** → correlation; **spread** → quartiles.

📈 VISUALIZATION One chart, one message - the statistic you quote is the chart you draw.

3 Who is the audience and what do they need?

Analyst? Manager? Executive? General public?

📊 STATISTICS Analysts want the full distribution; executives want one number in plain words.

📈 VISUALIZATION Match the detail to the decision - same finding, different chart and depth.

Question 1 - What type of data are you working with?

Understanding the variable types is crucial for selecting appropriate statistical methods and visualization techniques

Question 1 - Variable types 🔢

Five column types. Identify the column type first - it determines which statistics and which charts are appropriate.

Data type	Description & examples	Recommended statistics	Recommended visualizations
🏷 Categorical - Nominal	Categories without a natural order - region, payment method, product category.	Counts, proportions, mode	Bar chart, pie chart
🔢 Categorical - Ordinal	Categories with a meaningful order - Low/Medium/High, star ratings.	Counts, proportions, median, percentiles	Ordered bar chart
📏 Numeric	Measurable quantities - revenue, units sold, age.	Mean, median, standard deviation, range	Histogram, box plot, scatter plot
📅 Date-time	Time-based data points - order date, month, week.	Growth rate, period-over-period change	Line chart, time-series plot
✓ Boolean	Binary yes/no or true/false values - returned?, churned?, first-time buyer?	Rates, proportions	100% stacked bar, donut chart

Categorical Data

Qualitative (non-numeric) data that represents groups or categories

Nominal

Categories with no inherent order

Examples



Characteristics

- Distinct categories
- No natural order
- Labels or names (e.g., product type, country, animal)

Common charts



Ordinal

Categories with a meaningful order

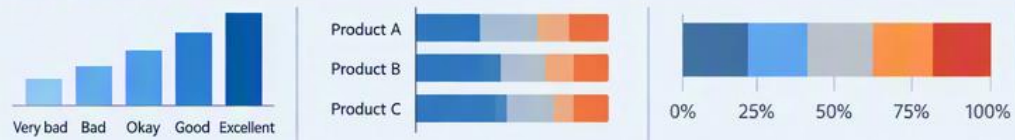
Examples



Characteristics

- Ordered categories (ranking)
- Can be sorted from low to high
- Common for ratings, levels, and preferences (e.g., satisfaction, education level, agreement)

Common charts



Categorical data describes groups or labels, not numbers.

Identify whether the data is nominal or ordinal to choose the right analysis and visualization.

Numerical Data

Quantitative (numeric) data that represents counts or measurements

Discrete

Numbers obtained by counting

Examples



Number of students



Number of orders



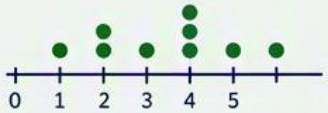
Number of cars



Number of defects/items

Key characteristics

- Countable values
- Usually whole numbers (integers)
- Gaps between possible values
- Cannot have fractions (e.g., 2.5 students or 3.7 cars)



Only specific values
(e.g., 0, 1, 2, 3, ...)

Common charts



Bar chart (counts)



Dot plot



Count plot

Continuous

Numbers obtained by measuring

Examples



Height (175.3 cm)



Weight (72.5 kg)



Temperature (36.8 °C)



Distance (12.7 km)

Key characteristics

- Measured values
- Can contain any real value (decimals)
- Infinite possible values within a range
- Precision depends on measurement

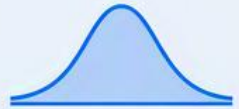


Any value within a range
(e.g., 12.7, 18.4, 25.0, ...)

Common charts



Histogram



Density plot



Box plot



Numerical data describes quantities and amounts.

Identify whether the data is discrete or continuous to choose the right analysis and visualization.

Question 2 - What do you want to communicate?

2 What do you want to **communicate**? - the statistics

COMPARISON



Used to compare values across groups or over time.

Common statistics:

- Mean and median per group
- Difference and % difference
- Growth rate over time
- Rank order of groups

COMPOSITION



Used to break a total down into its component parts.

Common statistics:

- Share of total (%)
- Counts per category
- Cumulative share
- Part-to-whole ratios

DISTRIBUTION



Used to describe how values are spread within one series.

Common statistics:

- Range, IQR, std. deviation
- Quartiles and percentiles
- Skew and outliers

RELATIONSHIP



Used to show whether two variables move together.

Common statistics:

- Correlation coefficient
- Cross-tabulation

💡 **Start simple:** the mean, median, a percentage and a growth rate answer most business questions. Use fancier measures only when these fall short.

2 What do you want to **communicate**? - the charts

COMPARISON



Used to compare values over time or across categories.

Common visuals:

- Bar chart
- Column chart
- Clustered column / bar
- Line or area chart (time)

COMPOSITION



Used to break down the component parts of a whole.

Common visuals:

- Stacked bar / column chart
- Pie or donut (3–4 parts)
- Stacked area chart
- Treemap or sunburst

DISTRIBUTION



Used to show the frequency of values within a series.

Common visuals:

- Histogram
- Box and whisker plot
- Density plot
- Violin plot

RELATIONSHIP



Used to show correlation between multiple variables.

Common visuals:

- Scatter plot
- Bubble chart
- Heat map
- Correlation matrix

💡 **Keep it simple!** Basic variations of bar, column and line charts are often the simplest - and clearest - way to visualize data.

Question 3 - Who is the audience and what do they need?

3 Who is the **audience** and what do they need?

ANALYST



Wants to reproduce and challenge your numbers.

Show them:

- Full distributions, box plots
- Scatter with every point
- Labelled, detailed axes
- Never hide outliers

MANAGER



Wants performance against a target, by team or region.

Show them:

- Sorted bars for clear ranking
- Actual vs. target with a target line
- Compare regions side by side

EXECUTIVE



Wants to decide quickly.

Show them:

- One to three KPIs
- A single annotated trend line
- One highlight colour
- No dense tables

GENERAL PUBLIC



Wants one memorable idea, with no prior context.

Show them:

- A simple bar or line chart
- Plain-language labels
- Rounded, familiar numbers
- Never truncate the y-axis



Ask before you build: what decision will this person make, what do they already know, and how many seconds will they really spend looking? Those three answers set the level of detail.

Statistics for Data Insights

Describing data with numbers: four topics, one workflow

1

Central Tendency

What single value is typical of the data?

2

Dispersion

How far do values spread around that centre?

3

Distribution Shape

Is the data symmetric, skewed or heavy-tailed?

4

Relationships

Do two variables move together?

Work in order: a centre is only meaningful once you state its spread, its shape, and the sample size n.

Learning outcomes

- Compute and interpret the mean, median, mode, range, variance, standard deviation, percentiles and IQR.
- Diagnose distribution shape with skewness and kurtosis, and detect outliers.
- Measure and correctly report the relationship between two variables.

1 Measures of Central Tendency

Choose and defend the **single number** that best represents a dataset.

Measures of Central Tendency

$\bar{x}$

Mean

The balance point of the data. Uses every value, so extreme values move it.

Best for symmetric data

$$\bar{x} = \Sigma x_i / n$$

M

Median

The middle value once the data is ordered.

Best for skewed data, or data containing outliers.

M = middle of ordered data

Mo

Mode

The most frequently occurring value. The only center defined for categories.

Best for categorical data and repeated discrete values

Mo = most frequent value

All three answer "what is typical value?"

In a perfectly symmetric distribution **mean = median = mode**.
They disagree whenever the distribution is not symmetric.

Worked Example: Five Trainees

Small enough to verify every statistic by hand

Name	Gender	Speaking /50	Math /50
Omar	M	40	38
Layla	F	45	41
Noura	F	23	42
Khalid	M	39	48
Yousef	M	39	32

Ordered speaking marks

23 • 39 • 39 • 40 • 45

Interpretation: a typical speaking mark is 39
The mean of 37.2 is dragged down by one low score.

Mean = 37.2

(40 + 45 + 23 + 39 + 39) ÷ 5 = 37.2 marks. Noura's 23 pulls it below every other mark.

Median = 39.0

Third value of five once ordered. Unaffected by how low the lowest mark is.

Mode = 39

39 occurs twice; every other mark occurs once. The most common mark.

Mean vs Median: The Effect of One Outlier

A controlled demonstration of what “robust” actually buys you

Before - clean data

Speaking: 40, 45, 23, 39, 39

Mean 37.2 • Median 39.0

Gap = 4.6% of the median - either center is defensible.

After - one typing error (45 → 450)

Speaking: 40, 450, 23, 39, 39

Mean 118.2 • Median 40.0

Gap = 195% - the mean is now a mark nobody scored.

Statistic	Before	After	Change	Why
Mean	37.2	118.2	+218%	Uses the magnitude of every value
Median	39.0	40.0	+2.6%	Uses rank only; one value moves it one position

Practical rule: Compute both. Investigate any gap larger than 10% of the median.

The Mode and Categorical Data

Payment method recorded for 200 orders

Value	Frequency	Proportion
Card	96	0.48
Cash	62	0.31
Wallet	30	0.15
Transfer	12	0.06
Total	**200**	**1.00**

Frequency and proportion

Frequency = number of times a value occurs.
Proportion = frequency ÷ sample size n.

Proportions always sum to 1 (100%).

Mode = “Card”

The most frequent value - the only measure of central tendency valid for categorical data.

Watch out

If all values occur with the same frequency, there is no meaningful mode.
On continuous data, group values into bins first.

Central Tendency in Pandas

Three methods, plus the split that makes them honest

PYTHON - CENTRE

```
df['speaking'].mean()      # 37.2
df['speaking'].median()    # 39.0
df['speaking'].mode()[0]   # 39
# alternative
df['speaking'].agg(['mean', 'median',
        lambda x: x.mode()[0]])
df.groupby('gender')['speaking'].mean()
# F  34.00   |   M  39.33
```

PYTHON - ONE-LINE SUMMARY

```
df[['speaking','math']].describe()
# count  mean   std   min   max
#  5.00  37.20  8.32  23.0  45.0
```

How to read the output

Row	Reading
count	5 - number of enntries
mean	37.20 - the balance point of all five marks.
median	39.00 - the median; compare it with the mean.
std	8.32 - average distance from the mean, in marks.
min / max	23 and 45 - the two values driving the range.

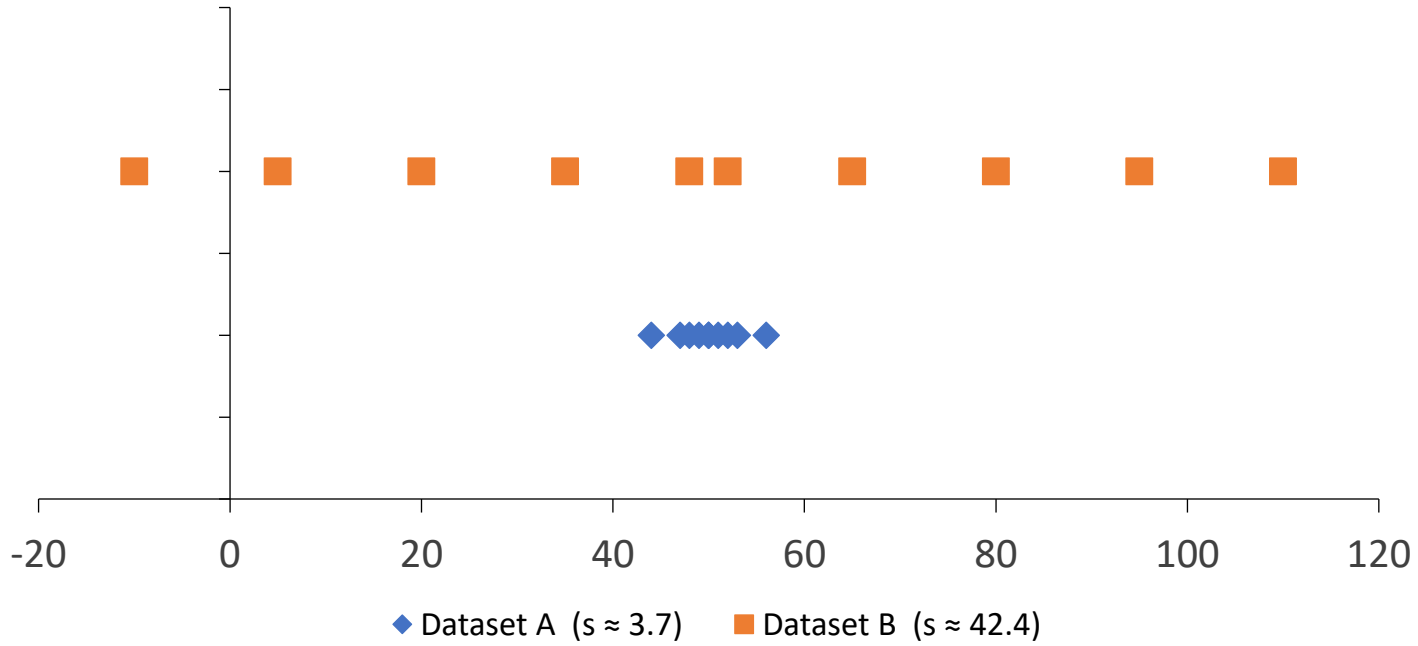
Always split before you summarise: one overall mean of 37.2 hides a 5.3-mark gap between groups (i.e., between genders)

2 Measures of Dispersion

Quantify how widely **data spreads**.

Why Spread Matters

Two datasets, the same mean, completely different behaviour



Same centre

Both datasets have a mean of 50.
A center alone cannot distinguish them.

Different risk

Dataset A is predictable, while Dataset B ranges from −10 to 110.
The same decision may be safe for Dataset A but risky for Dataset B.

Dispersion tells you how well the center represents the data: how much the values vary around it.

Four standard measures: range, variance, standard deviation and the interquartile range (IQR).

Range, Variance and Standard Deviation

Measuring the average distance from the mean

Range

Largest value minus smallest. Easy to compute, but based on only two observations and grows with sample size.

$$R = x_{max} - x_{min}$$

Variance s^2

Mean of the squared deviations from $\bar{x}$. Squaring stops positive and negative deviations cancelling - but the units become “marks squared”.

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

Standard deviation s

Square root of the variance, so it returns to the original units. This is the figure you quote beside the mean.

$$s = \sqrt{s^2}$$

Speaking marks (n = 5)	Range = 22	s^2 = 69.2 marks^2	s = 8.32 marks
Report it as	context only	for calculation only	“mean 37.2 ± 8.3 marks”

Divide by n − 1, not n: it corrects the bias that arises when a sample is used to estimate a population.

Percentiles, Quartiles and the IQR

Definition

The pth percentile is the value below which p% of the observations fall.

IQR = Q3 – Q1

The width of the middle 50% of the data

Why the IQR is robust: It ignores the extreme quarter at each end, so single extreme values cannot inflate it.

Statistic	Meaning
Q1 (25th pct)	25% of values lie below it
Q2 (50th pct)	The median
Q3 (75th pct)	75% of values lie below it
IQR	Spread of the central half

Use the pair that matches the shape: mean ± s for symmetric data, median + IQR for skewed data.

Worked example of 2,400 orders

Q1	Median	Q3	IQR
142	268	497	355

Reading the asymmetry

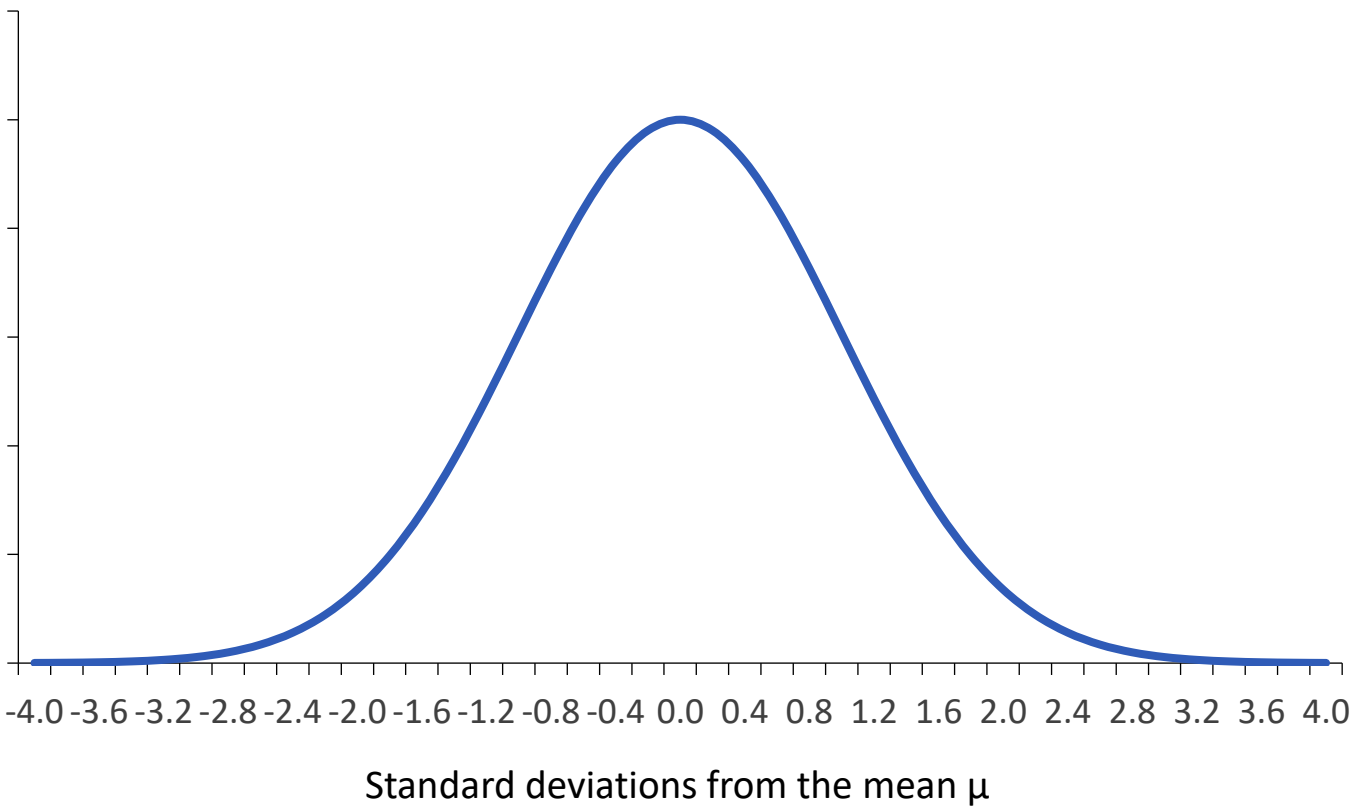
Q3 is 229 units above the median, while Q1 is only 126 units below it → indicating greater spread on the upper side and suggesting a longer right tail.

- If Q3 − Q2 > Q2 − Q1 (Q2 closer to Q3 than Q1)→ suggests a longer right tail and possible positive skewness.
- If Q2 − Q1 > Q3 − Q2 (Q2 closer to Q1 than Q3) → suggests a longer left tail and possible negative skewness.
- If the distances are approximately equal, the distribution may be roughly symmetric.

Note: This is a useful indication, examine the overall distribution or a box plot to confirm.

The Normal Distribution and the 68–95–99.7 Rule

For normally distributed data, most observations are close to the mean.



$\mu \pm 1s \rightarrow 68.3\%$

About 68% of observations fall within one standard deviation of the mean.

$\mu \pm 2s \rightarrow 95.4\%$

About 95% of observations fall within two standard deviations of the mean.

$\mu \pm 3s \rightarrow 99.7\%$

About **99.7% of observations** fall within three standard deviations of the mean. Only **0.3%, or about 3 in 1,000**, fall outside this range.

Coefficient of Variation

Comparing spread across different scales and units

$$CV = s / \bar{x}$$

Unitless, so it compares columns measured on different scales

CV	Reading
< 0.3	Consistent - mean and s describe the column well
0.3 – 1.0	Variable - report both centre and spread
> 1.0	Spread exceeds the average - stop using mean and s

Never compute the CV when the mean is near zero or can be negative (temperatures, profit).

Worked comparison

Column	Mean	s	CV
Speaking /50	37.2	8.32	0.22
Math /50	40.2	5.85	0.15
Order value QAR	418.60	512.30	1.22

Interpreting the CV

- **Speaking:** CV = **0.22 (22%)**
  - Scores vary by about **22% of the mean**.
  - Speaking scores are **less consistent** than math scores.
- **Math:** CV = **0.15 (15%)**
  - Scores vary by about **15% of the mean**.
  - Math scores are **more consistent** than speaking scores.
- **Order Value:** CV = **1.22 (122%)**
  - The standard deviation is **larger than the mean**.
  - The mean alone is not very representative; the **median and IQR** may provide a better summary.

3 Measures of Distribution Shape

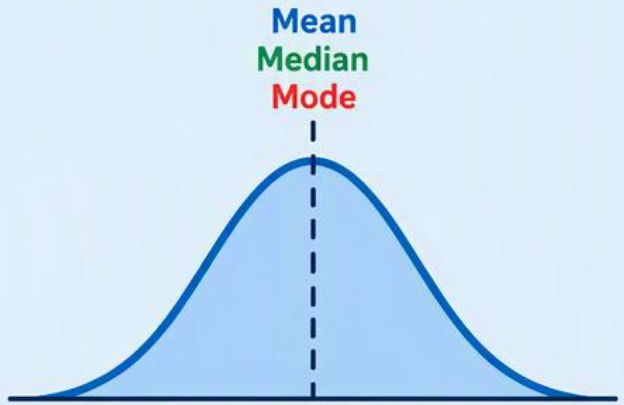
Read the shape of a distribution using **Skewness and Kurtosis** and decide which statistics remain valid.

Mean, Median and Mode in Distributions

The shape of a distribution affects the relationship between the mean, median and mode.

Normal (Symmetric) Distribution

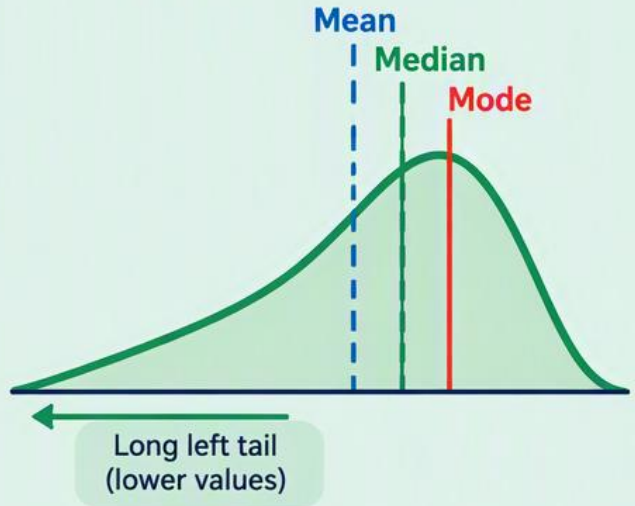
In a normal symmetric distribution, the mean, median and mode are the same.



Symmetric shape
(same on both sides)

Left-Skewed (Negative-Skewed) Distribution

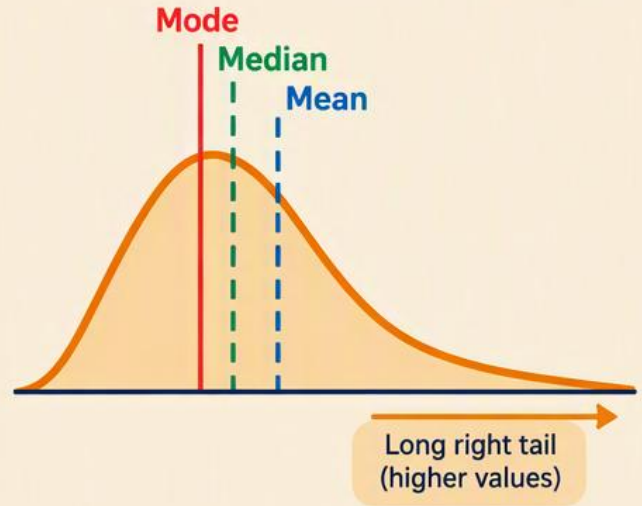
A left-skewed (negative-skewed) distribution has a long left tail (the tail on the lower values).



Mean < Median < Mode

Right-Skewed (Positive-Skewed) Distribution

A right-skewed (positive-skewed) distribution has a long right tail (the tail on the higher values).



Mode < Median < Mean

Key Takeaway:

A distribution is skewed if one of its tails is longer than the other.
The position of the mean, median and mode helps us understand the shape and direction of the skewness.

The Coefficient of Skewness

Quantifying asymmetry around the mean

$$g_1 = (1/n)\ \Sigma(x_i - \bar{x})^3 / s^3$$

Cubing preserves the sign, so the direction of the tail survives

Skewness	Shape	Report
< −0.5	Left tail	Median
−0.5 to +0.5	Roughly symmetric	Mean
> +0.5	Right tail	Median

PYTHON

```
df['order_amount'].skew()     # +4.12
```

Worked example

- **Order Value: Skewness = +4.12 → Strongly right-skewed.** Most orders are relatively small, but a few very large orders create a long right tail. This pattern is common for **revenue, income, insurance claims, and transaction amounts**.
- **Speaking Marks: Skewness = −1.70 → Left-skewed (negatively skewed).** Most students achieved relatively high scores, while a few low scores pull the distribution to the left. This is consistent with **Mean = 37.2 being lower than Median = 39.0**.

- **Standard remedy:** Apply a **log transformation** to compress the large values in a long right tail. For example, **log1p(order_value)** reduces the skewness from **4.12** to **0.21**, making the distribution **approximately symmetric**.
- **Analyze the data on the log scale, but report results on the original scale** (e.g., **QAR)** so they remain easy to interpret.

Kurtosis: How Common Are Extreme Values?

Kurtosis measures how often unusually large or small values occur compared to a normal distribution.

$$g_2 = (1/n)\,\Sigma(x_i - \bar{x})^4 / s^4 - 3$$

pandas returns excess kurtosis: 3 is already subtracted, so 0 means normal-like tails

$g_2 \approx 0$

Normal-like tails

- Extreme values occur about as often as a normal distribution.
- The **68-95-99.7 rule** is a reasonable approximation.

$g_2 > 0$ (heavy tails)

- Extreme values occur **more often** than normal.
- Outliers are more common and can strongly affect the mean and standard deviation.

$g_2 < 0$ (light tails)

Light tails

- Extreme values occur **less often** than normal.
- Most observations are concentrated near the center.

- Skewness tells us the direction of the tail.
- Kurtosis tells us how heavy the tails are and how likely extreme values are.

Example: Order Value kurtosis = **26.8**

- Extreme order values are much more common than in a normal distribution.
- Investigate outliers and consider robust measures such as the **median and IQR**.

Detecting Outliers

An outlier is a value that is unusually far from the rest of the data -> an unusual value, not necessarily an error.

Method 1: Z-Score Rule (approximately symmetric data)

$$z = \frac{x - \bar{x}}{s}$$

Use when the data is roughly normal or symmetric.

Method 2: IQR Rule (skewed data)

Use when the data is skewed or contains extreme values.

Outlier if |z| > 3

Why? For a normal distribution, about **99.7%** of values lie within **±3 standard deviations** of the mean.

Flag as a potential outlier if:

x < Q1 – 1.5·IQR or x > Q3 + 1.5·IQR

Orders data	**Q1 = 142**	**Q3 = 497**	**IQR = 355**	**Upper fence = 1,029.50**
Result	118 of 2400 orders (4.9%) exceed the upper fence.	Normal data would give ≈ 0.7%	The excess measures skew	This suggests a strong right tail (skewness).

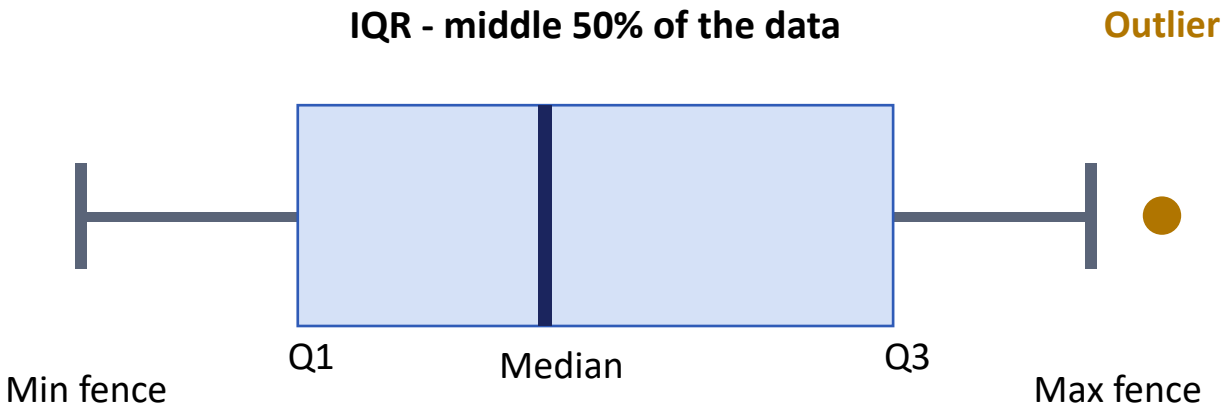
A fence flags values that are far from the middle.

⚠️ An outlier is a value worth investigating, not automatically a error.

e.g., The flagged orders are unusual, not necessarily errors.

Boxplots: Reading Spread and Outliers at a Glance

One chart showing center, spread, skew and extremes together



Whiskers extend to the furthest value within 1.5 × IQR of the box; points beyond them are plotted individually.

What to read first

Where the median line sits inside the box. Off-centre means the distribution is skewed.

Why it beats a table

Centre, spread, asymmetry and extreme values are visible in one glance - and several groups can be compared side by side.

Box height = IQR • line inside the box = median • whiskers = 1.5 × IQR • separate points = flagged outliers

In pandas: `df.boxplot(column='price', by='region')` - one call to compare the distribution across groups.

Handling Outliers: Investigate Before You Remove

1 **Detect**

Using Z-score rule for Symmetric data and IQR rule for Skewed data

2 **Investigate**

- Is it a data-entry or measurement error?
- Is it a valid but rare observation?
- Does it represent a different group?

Example: Luxury properties may appear as outliers in housing-price data.

3 **Decide Based on the Goal**

For Data Analysis

✅ Investigate and understand outliers.

✅ Keep and analyze them separately if they represent a meaningful subgroup.

Example: Analyze luxury properties separately from the general housing market rather than removing them.

For Machine Learning

✅ Outliers may be: Corrected (if erroneous), Removed (if invalid), Capped (Replace values beyond the IQR fences with the corresponding fence values).

This is often necessary because some algorithms, such as **Linear Regression**, are sensitive to extreme values.

⚠️ **Never remove outliers automatically. The reason they are unusual may be the most important finding in the data.**

4 Analysis of Relationships Between Variables

Measure how two variables move together.

What It Means for Two Variables to Be Related

Association

- **Association:** Knowing one variable gives us useful information about the likely value of the other.
- **What it tells us:** It shows a **pattern in the data**, not that one variable causes the other.

Causation

- Causation means that changing one variable causes a change in another.
- A controlled experiment is usually needed to confirm causation.

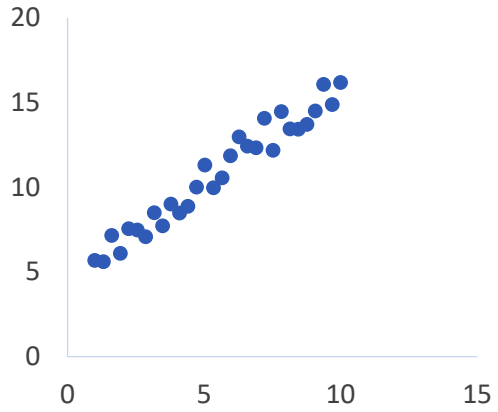
A realistic example: a retail chain's monthly data

- **Observed:** Months with higher advertising spend also had higher revenue. This shows an **association**.
- **Not proven:** We cannot say that higher advertising **caused** higher revenue. Seasonality may have increased both.
- **Test it:** Keep advertising spend unchanged for similar stores and compare their revenue. A controlled comparison provides stronger evidence of **causation**.

Always Start with a Scatterplot

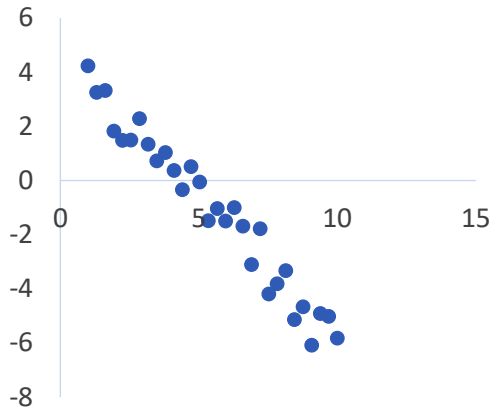
One point per observation - x is one variable, y is the other

Positive



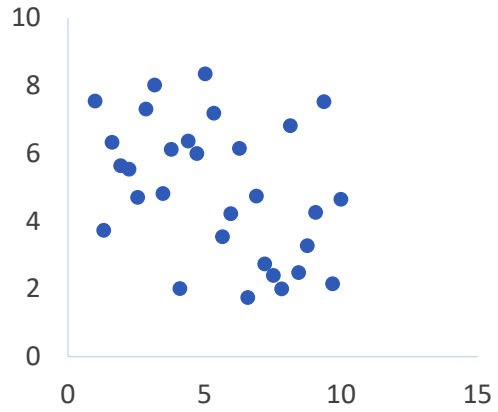
As ad spend rises, revenue rises.

Negative



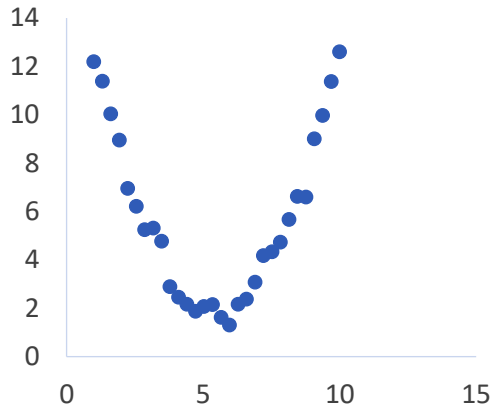
As discount deepens, margin falls.

Weak / none



Email sends tell us little about returns.

Non-linear



Price vs. units sold: strong, but curved.

Read four things in this order: Direction (up, down, neither) → **Form** (straight or curved) → **Strength** (how closely the points follow the pattern) → **Unusual points** (look for outliers or unusual patterns).

Pearson's Correlation Coefficient

The strength of a linear relationship between two continuous variables

$$r = \Sigma(x-\bar{x})(y-\bar{y}) \,/\, \sqrt{[\Sigma(x-\bar{x})^2 \cdot \Sigma(y-\bar{y})^2]}$$

r	Interpretation
+1.0	Perfect positive linear relationship
+0.3 to +0.7	Moderate positive
–0.3 to +0.3	Weak or none
–0.7 to –1.0	Strong negative

PYTHON

```
df[['price','sales']].corr()
```

How to read the sign

- Positive r: as one variable increases, the other tends to increase.
- Negative r: as one increases, the other tends to decrease.
- **r = 0 means no LINEAR relationship**

When Pearson's r Is Valid

Three assumptions - and what to do when they fail

1

Linearity

The relationship must follow a straight line. A curved pattern can give r ≈ 0 even when the variables are strongly related.

2

Normality

Both variables should be approximately normally distributed, without heavy tails or extreme values.

3

Independence

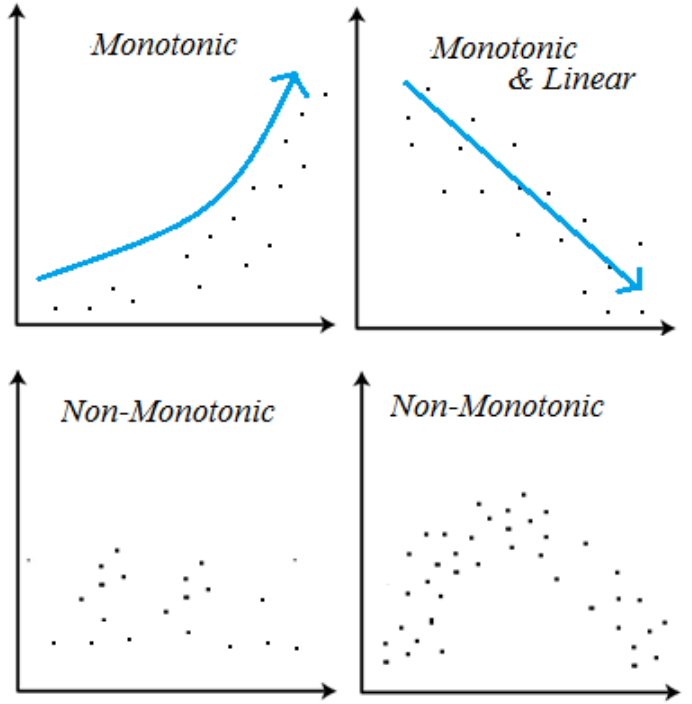
No **repeated measurements** of the same person, store, customer, or other unit.

If this fails	Symptom	Do this instead
Linearity	Pearson and Spearman disagree sharply	Use Spearman, or transform the variable
Normality	Strong skew or outliers present	Use Spearman's rank correlation
Independence	Repeated observations	r is not valid

Always plot the scatter before reporting r - a single number cannot reveal a curve.

Spearman's Rank Correlation

- Spearman's Rank Correlation tells us **how consistently two variables move together**, based on their ranked positions rather than their actual values.
  - **Monotonic:** The two variables move in the *same general direction* — when one increases, the other tends to increase consistently.
  - **Linear Monotonic:** The variables move in the same direction at an approximately constant rate (straight-line trend).
  - **Non-Monotonic:** The variables *do not follow any consistent direction* — the relationship goes up and down unpredictably
  - **Non-parametric:** does not assume a specific distribution of the data



Students	Maths	Science
A	35	24
B	20	35
C	49	39
D	44	48
E	30	45

Students	Maths	Rank	Science	Rank	d	d square
A	35	3	24	5	2	4
B	20	5	35	4	1	1
C	49	1	39	3	2	4
D	44	2	48	1	1	1
E	30	4	45	2	2	4
						14

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

ρ = Spearman's rank correlation coefficient

d_i = difference between the two ranks of each observation

n = number of observations

1 - (6 * 14) / 5(25 - 1) = 0.3

The Spearman's Rank Correlation for the given data is 0.3. The value is near 0, which means that there is a weak correlation between the two ranks.

How to Interpret Spearman's ρ

- **Direction:**

  ρ > 0: As one variable increases, the other tends to increase.

  ρ < 0: As one variable increases, the other tends to decrease.
- **Strength:** Use **|ρ|** (absolute value): **Closer to ±1 = stronger relationship vs. Closer to 0 = weaker relationship**
  - **0.00–0.19** → Very weak / none
  - **0.20–0.39** → Weak
  - **0.40–0.59** → Moderate
  - **0.60–0.79** → Strong
  - **0.80–1.00** → Very strong
- **Key point:** Spearman's ρ measures the strength and direction of a monotonic relationship based on ranks.
  - Unlike Pearson's correlation, the relationship does not need to be linear.
- **Example:** ρ = **–0.72** → **strong negative** relationship: higher values of one variable tend to correspond to lower values of the other.

The Correlation Matrix: Every Pair at Once

RetailCo monthly data, 36 months - five numerical variables

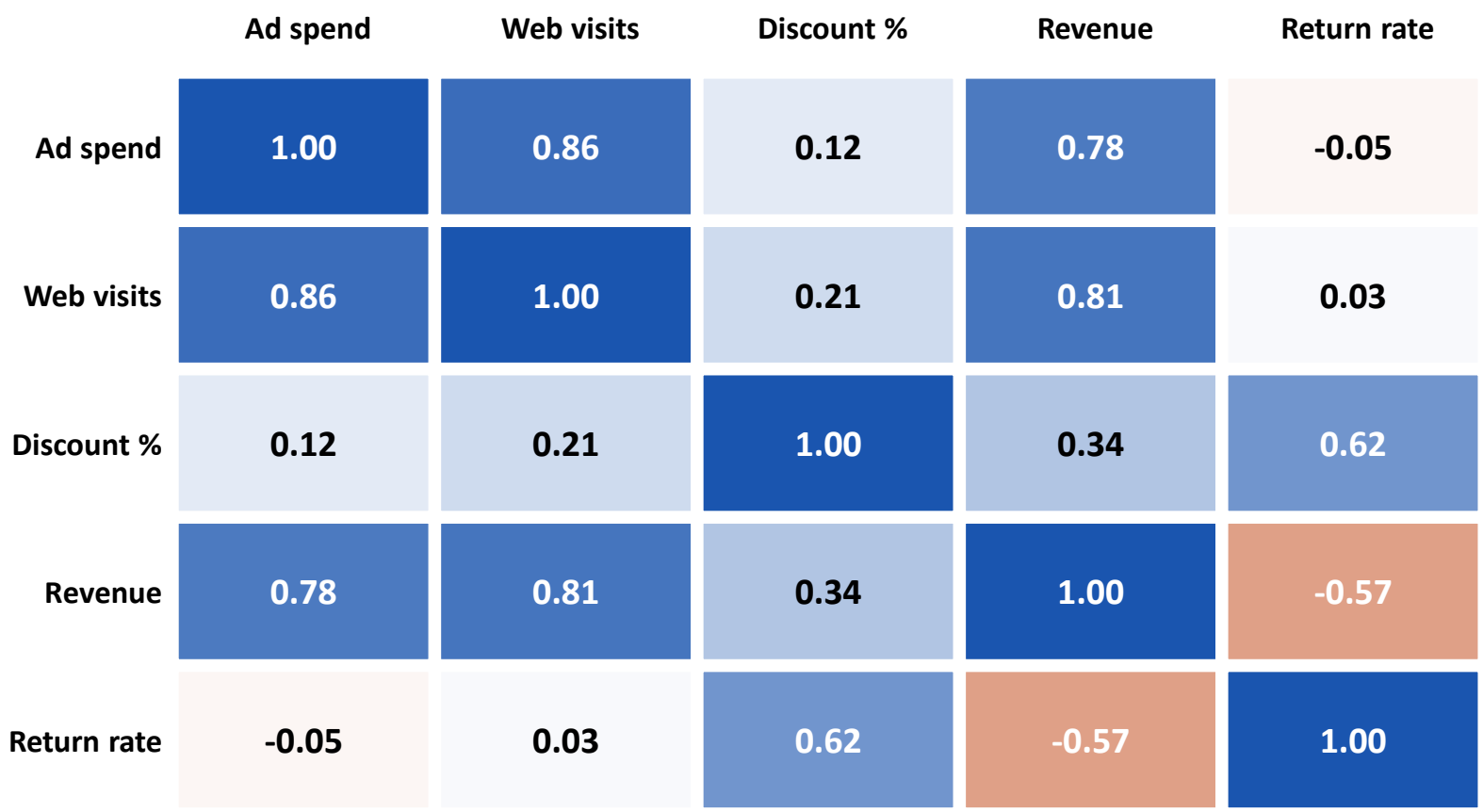
	Ad spend	Web visits	Discount %	Revenue	Return rate
Ad spend	1.00	**0.86**	0.12	0.78	-0.05
Web visits	0.86	1.00	0.21	0.81	0.03
Discount %	0.12	0.21	1.00	0.34	**0.62**
Revenue	0.78	0.81	0.34	1.00	-0.57
Return rate	-0.05	0.03	0.62	**-0.57**	1.00

What stands out here

- **Strongest positive:** ad spend ↔ web visits, 0.86.
- **Strongest negative:** revenue ↔ return rate, −0.57.
- **Worth investigating:** discount % ↔ return rate, 0.62 - deeper discounts come back more often.
- **Ignore for now:** ad spend ↔ return rate, −0.05.

The Correlation Heatmap: The Same Matrix, Read Faster

Colour encodes the coefficient so the eye finds the pattern



Dark blue = strong positive · white = near zero · dark orange = strong negative

Reading it

What do we see? A dark block linking ad spend, web visits and revenue; an orange cell at revenue × return rate.

What does it mean? Spend and traffic move together with revenue; months with more returns earned less.

What insight? Discounting (0.62 with returns) may be buying sales that come back.

What action? Test a capped-discount policy and monitor returns and net revenue.

In pandas: `sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm', center=0)`

Section Recap: The Analysis Checklist

Step	Question	What to Check	Practical Rule
1	What is a typical value?	Mean, Median, Mode	If mean and median differ substantially, report the median.
2	How much variation is there?	Standard Deviation (s), IQR, CV	If CV > 1, prefer Median and IQR over Mean ± s.
3	What is the shape?	Skewness, Kurtosis	Skewness: direction of asymmetry. If |skew| > 1, the data is strongly skewed. Kurtosis: frequency of extreme values.
4	Are there outliers?	IQR fences, Z-scores	Flag outliers and investigate before removing.
5	What service level can we promise?	P90, P95, P99	Use percentiles, not the mean.
6	Are variables related?	Correlation, Scatter Plot	Report correlation and always inspect a scatter plot.
7	Does the relationship hold for subgroups?	Group-by Analysis	If conclusions change across groups, report the subgroup results.
8	What can we conclude?	Describe what you observed, and acknowledge limitations	Report trends, differences, and associations supported by the data.

Recommended Analysis Flow

Center → Spread → Shape → Outliers → Relationships → Subgroups → Conclusions.